ANTI-INFLAMMATORY DRUGS

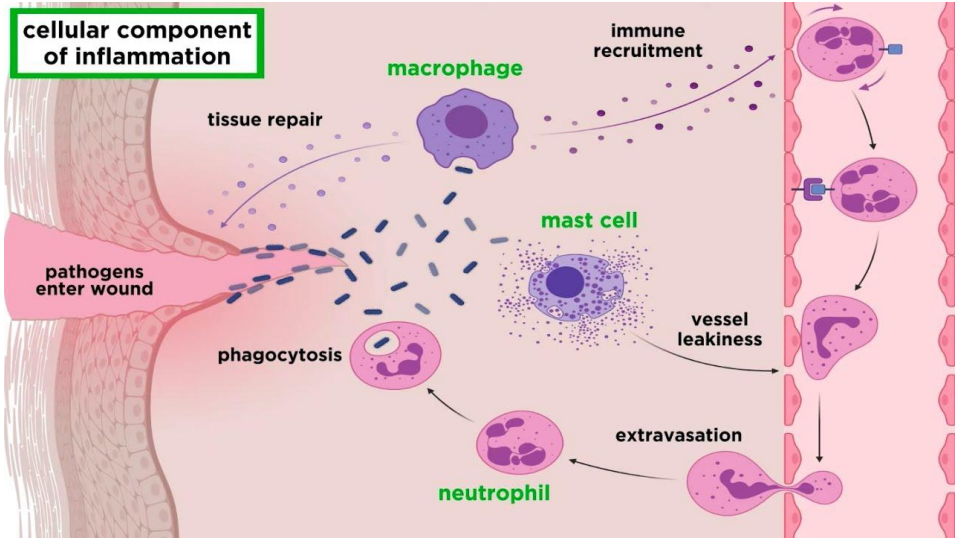
Isaac Edron J. Orig, RMT, MD
APMed-1

Inflammation

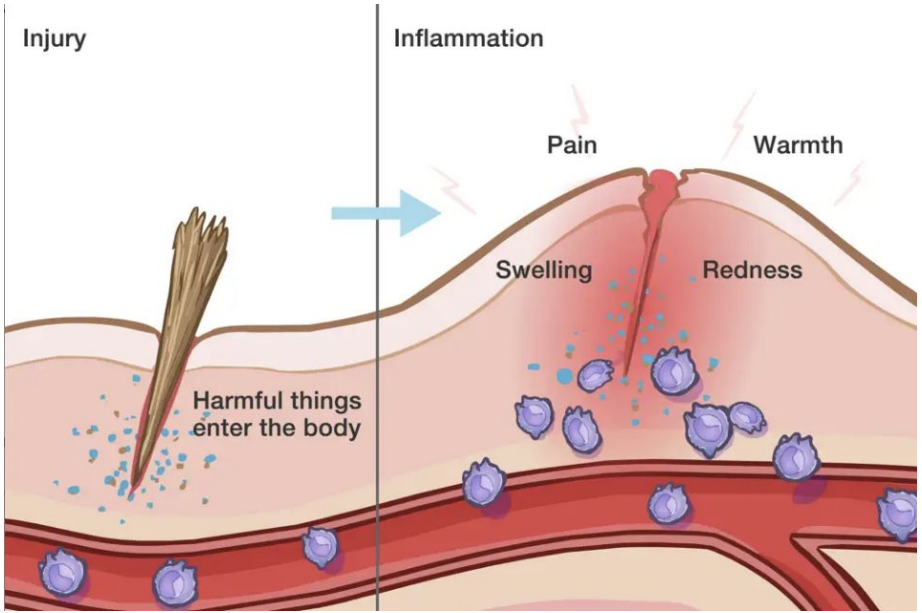
Inflammation

The immune system’s protective response to an injurious stimulus

Can be evoked by noxious agents, infections, and physical injuries



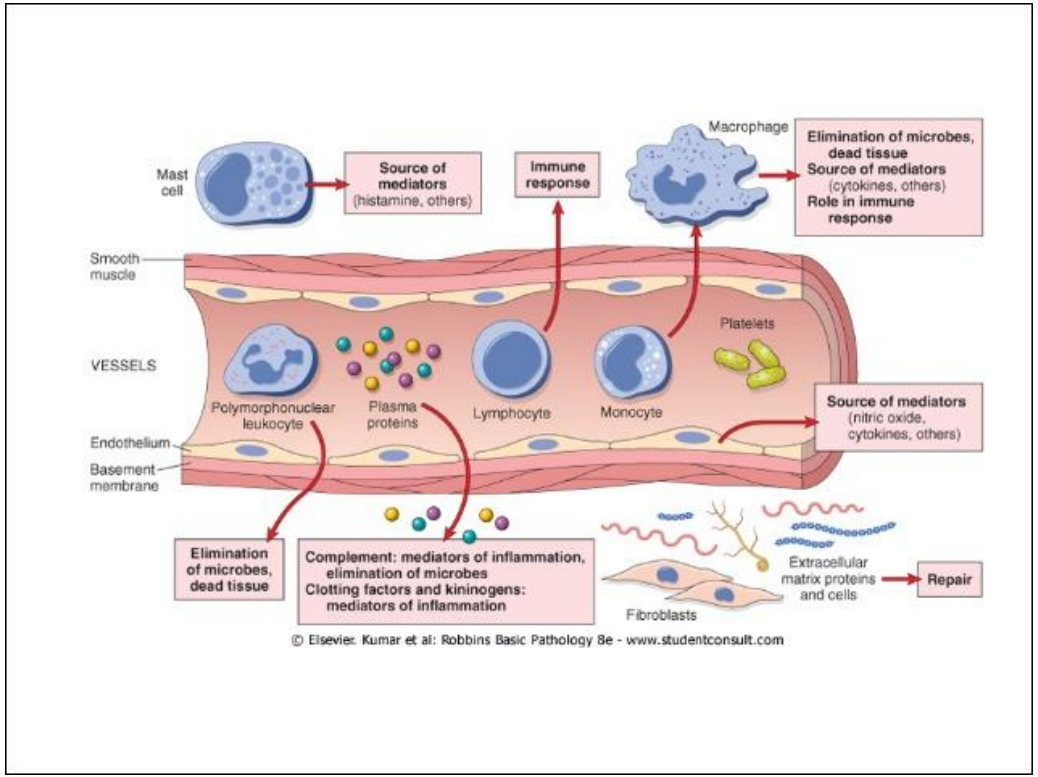


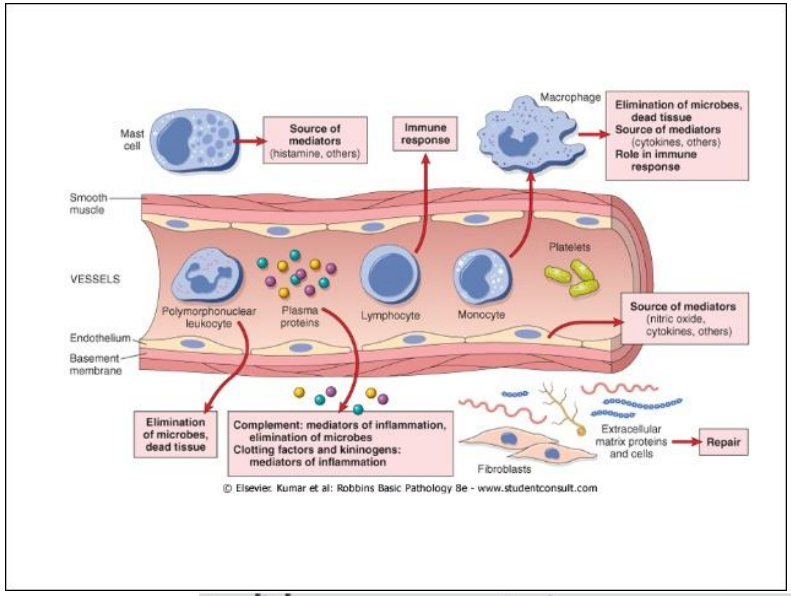

Injury
Inflammation
Pain
Warmth
Swelling
Redness
Harmful things
enter the body

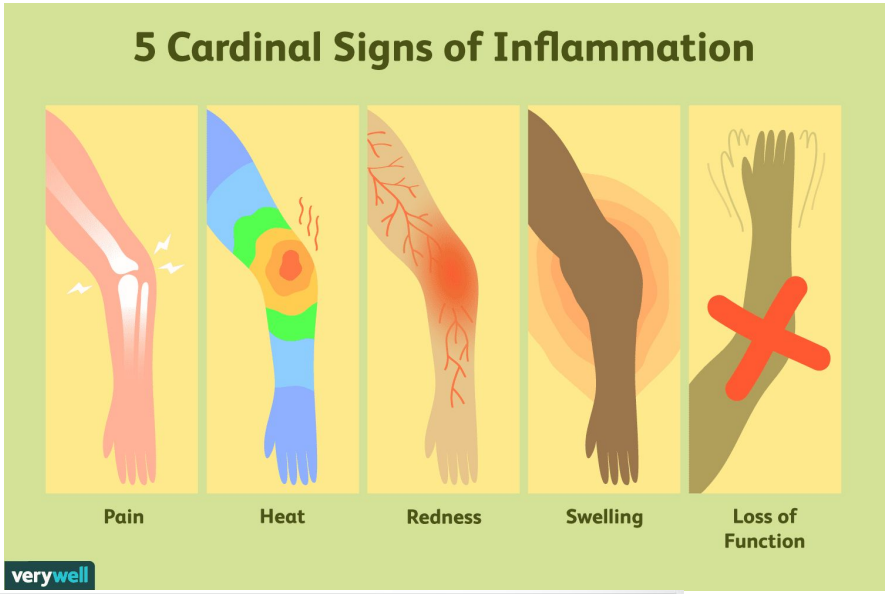
Inflammation

Characterized by:

1. Transient **local vasodilation** and **increased capillary permeability**
2. Infiltration of **leukocytes and phagocytic cells**
3. **Resolution** with or without tissue degeneration and fibrosis





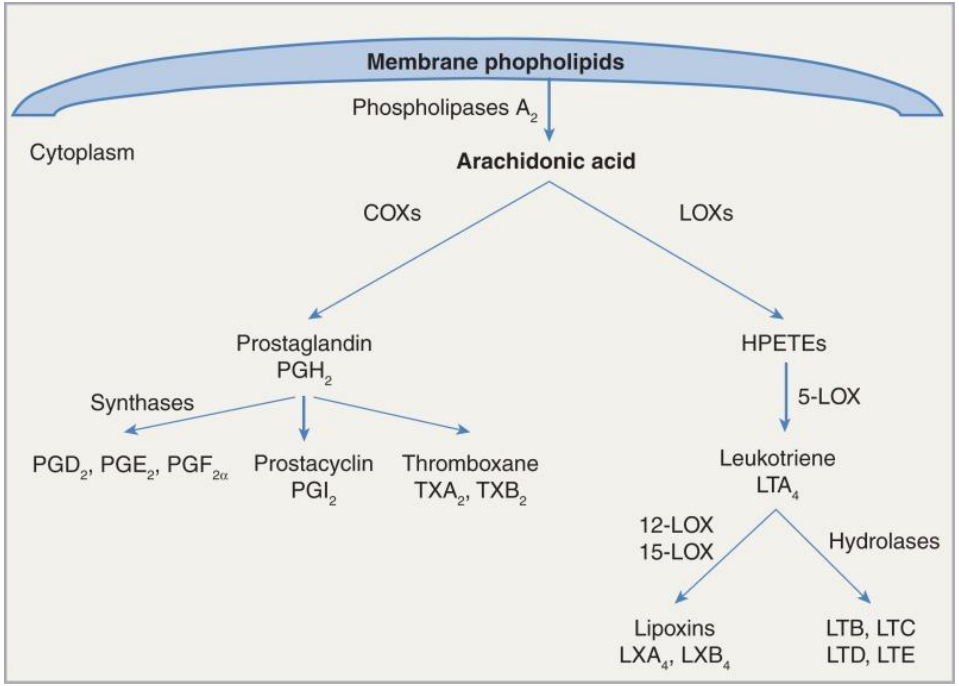


English	Latin	Cause
Heat	*Calor*	Vasodilation
Redness	*Rubor*	Vasodilation
Swelling	*Tumor*	Increased vascular permeability
		Increased granulation tissue
Pain	*Dolor*	Physical and chemical stimulation of nociceptors
Loss of function	*Functio laesa*	Pain
		Reflex muscle inhibition
		Disruption of tissue structure

Inflammation

Prostanoid synthesis is significantly increased

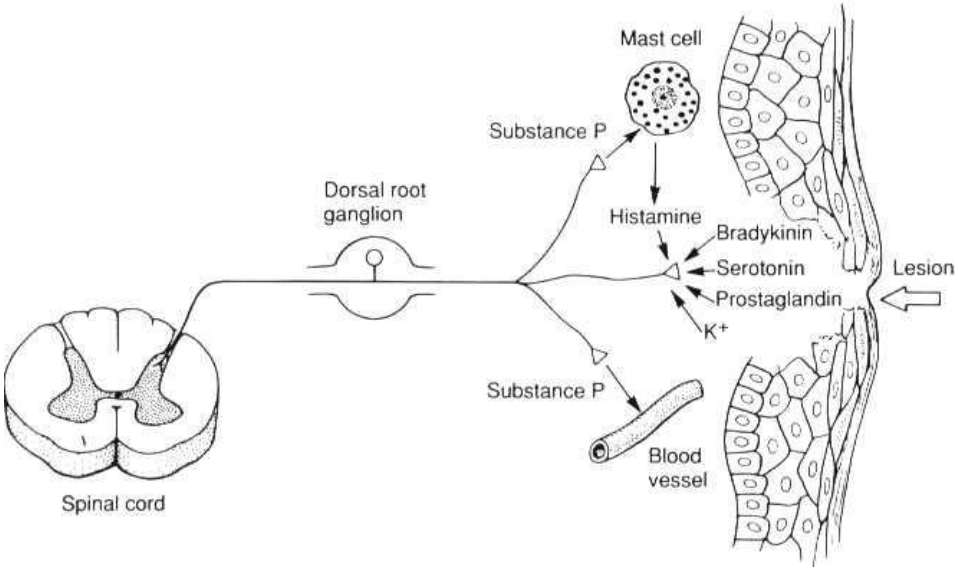
1. **PGE2, PGI** - primary prostanoids that mediate inflammation
   - increase local blood flow, vascular permeability and leukocyte infiltration
2. **PGD2** - allergic responses



Pain

Nociceptors are nerve fibers that sense pain

- activated by various stimuli such as heat, acid, or pressure



Inflammatory mediators increase sensitivity of nociceptors (PGs)

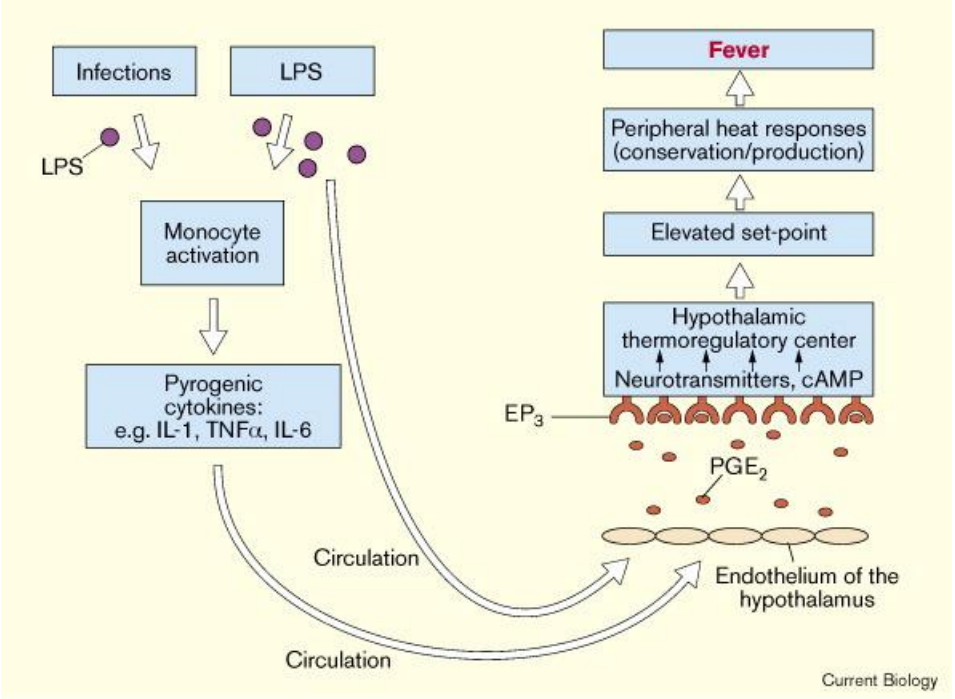
Fever

Hypothalamus regulates the set point at which the body temperature is maintained

- elevated in fever

Pyrogens

- IL-1, IL-6, TNF-alpha
- PGE2

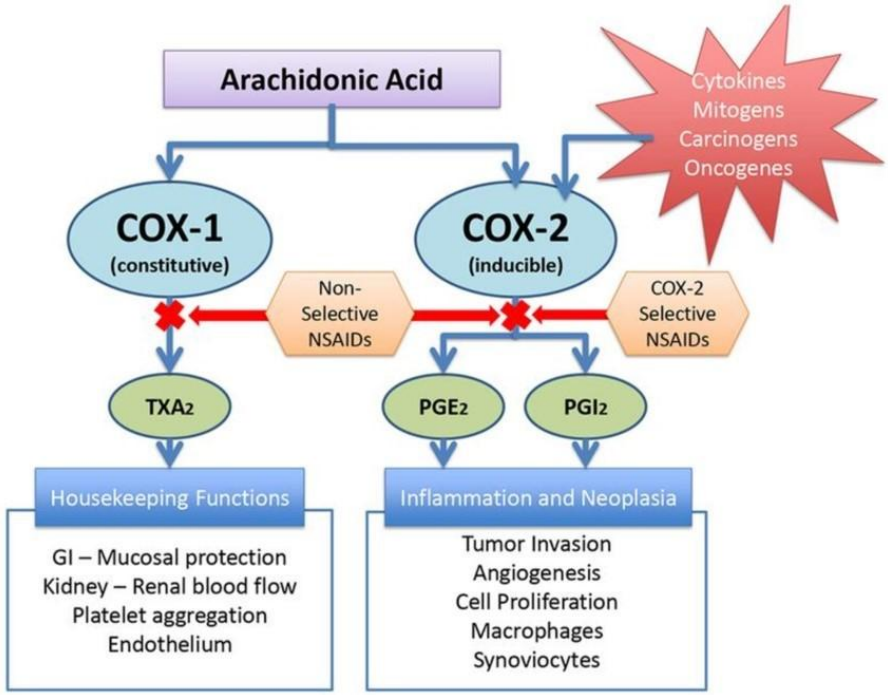


NSAIDs

Nonsteroidal Anti-inflammatory Drugs (NSAIDs)

Classified as:

1. Nonselective NSAIDs (both COX-1 and COX-2)
2. COX-2-selective NSAIDs



NSAIDs Mechanism of Action

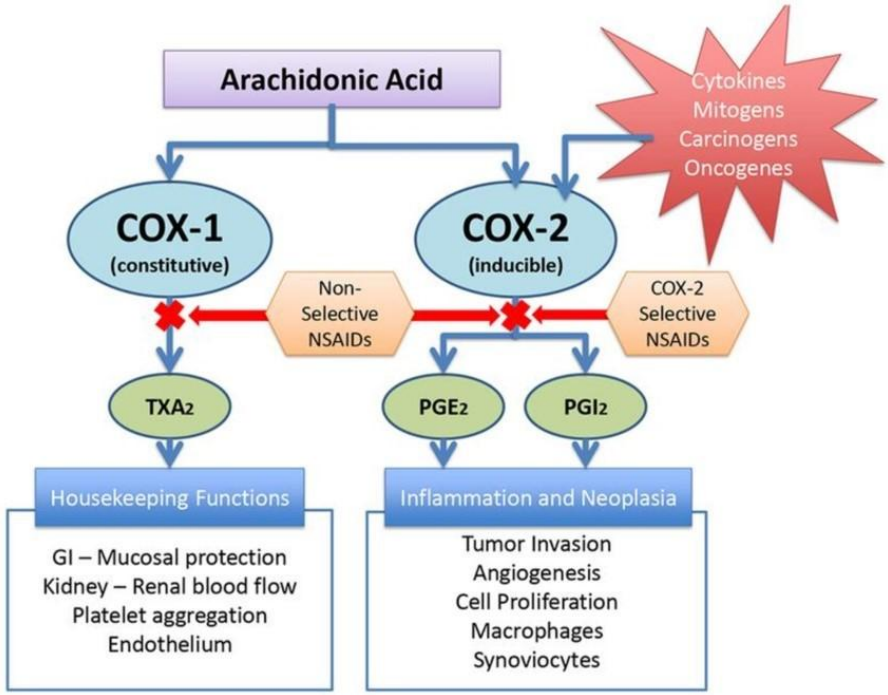
Principal effects are due to inhibition of PG production

COX-1

- expressed constitutively
- used for housekeeping functions

COX-2

- source of prostanoids for inflammation and cancer



Therapeutic Uses

1. Inflammation
2. Pain
3. Fever

Therapeutic Uses

Inflammation

- symptomatic relief from pain associated with musculoskeletal disorders (RA, osteoarthritis, gout)

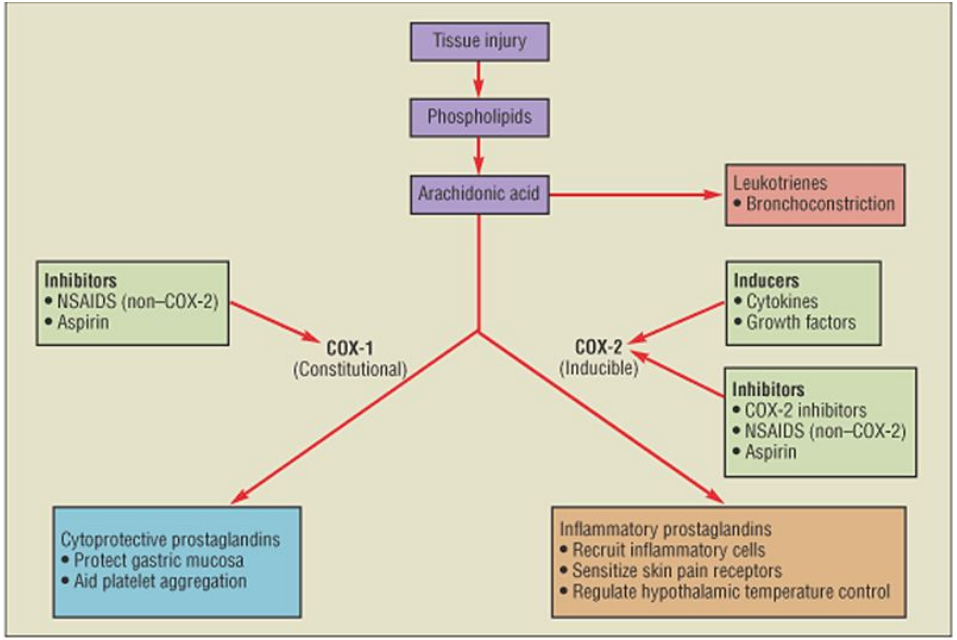


FIGURE 1. Algorithm of the biochemical pathway shows that the formation of prostaglandins occurs via both cyclooxygenase enzymes (COX-1 and COX-2).



Therapeutic Uses

Pain

- effective against inflammatory pain of low-to-moderate intensity
- can be used to treat primary dysmenorrhea and migraine attacks

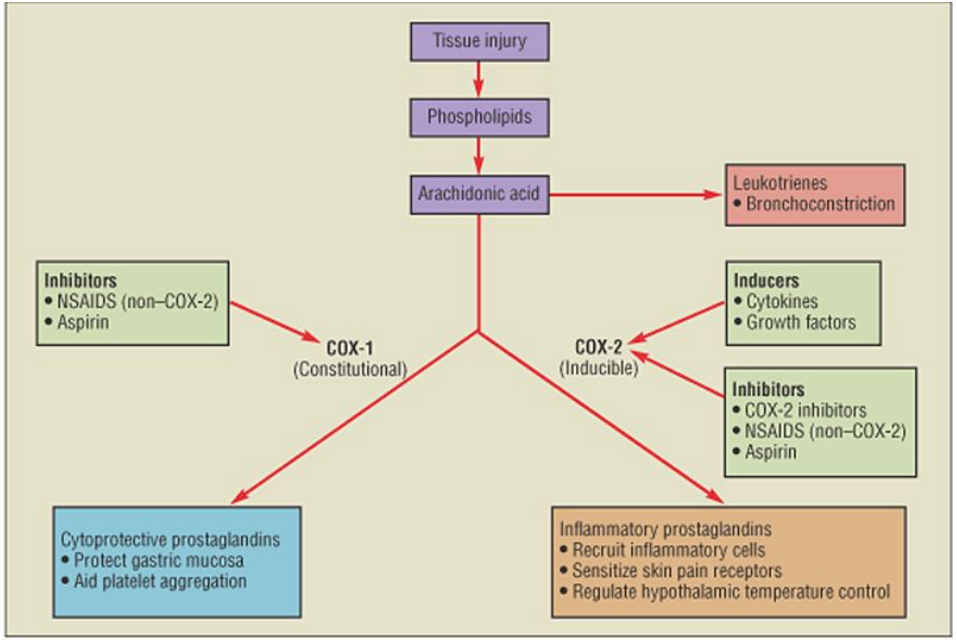


FIGURE 1. Algorithm of the biochemical pathway shows that the formation of prostaglandins occurs via both cyclooxygenase enzymes (COX-1 and COX-2).

Therapeutic Uses

Fever

- reserved for patients whom fever may be deleterious
- for relief when fever is lowered

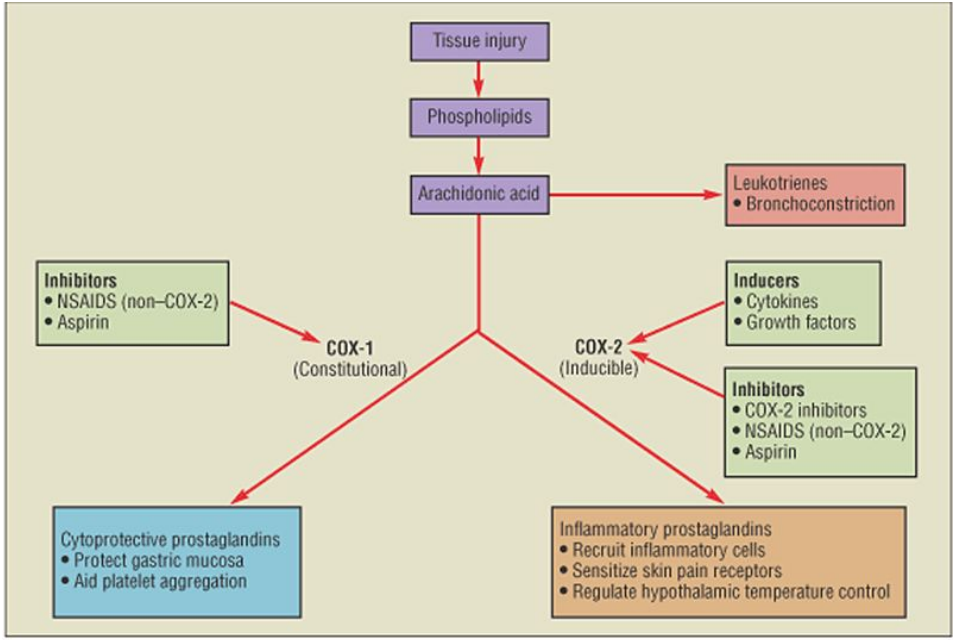


FIGURE 1. Algorithm of the biochemical pathway shows that the formation of prostaglandins occurs via both cyclooxygenase enzymes (COX-1 and COX-2).

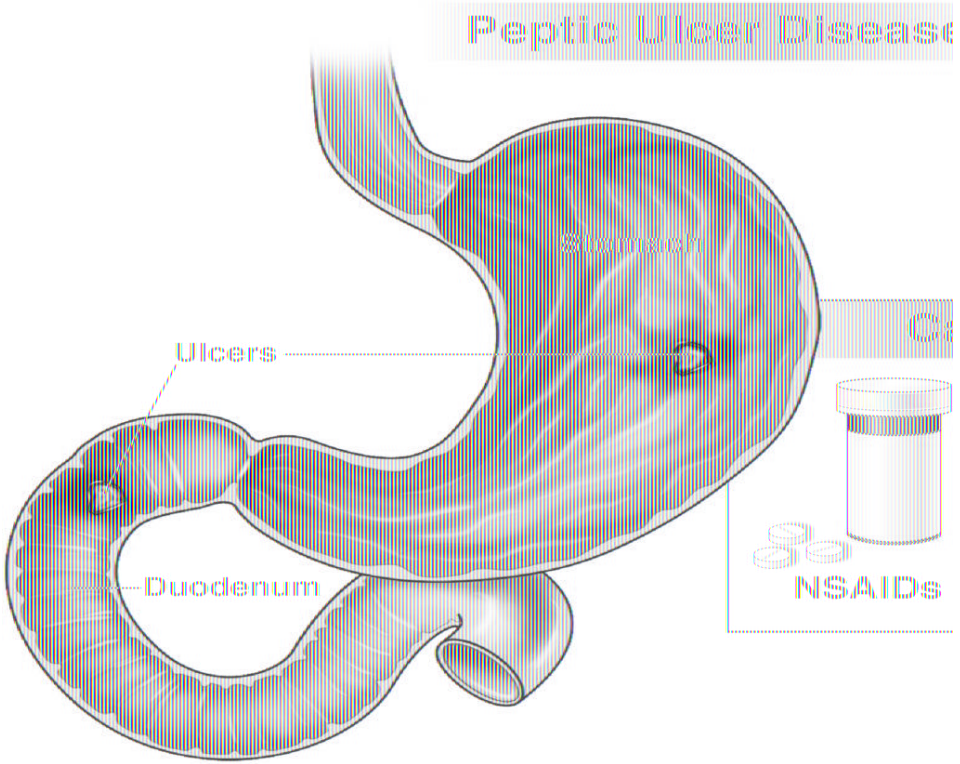
Adverse Effects

1. Gastrointestinal
2. Cardiovascular
3. Blood Pressure and Renal

Adverse Effects

Gastrointestinal

- most common symptoms associated with NSAIDs are GI (40%)
- may be serious (bleeding, perforation, obstruction) (1%-2%)



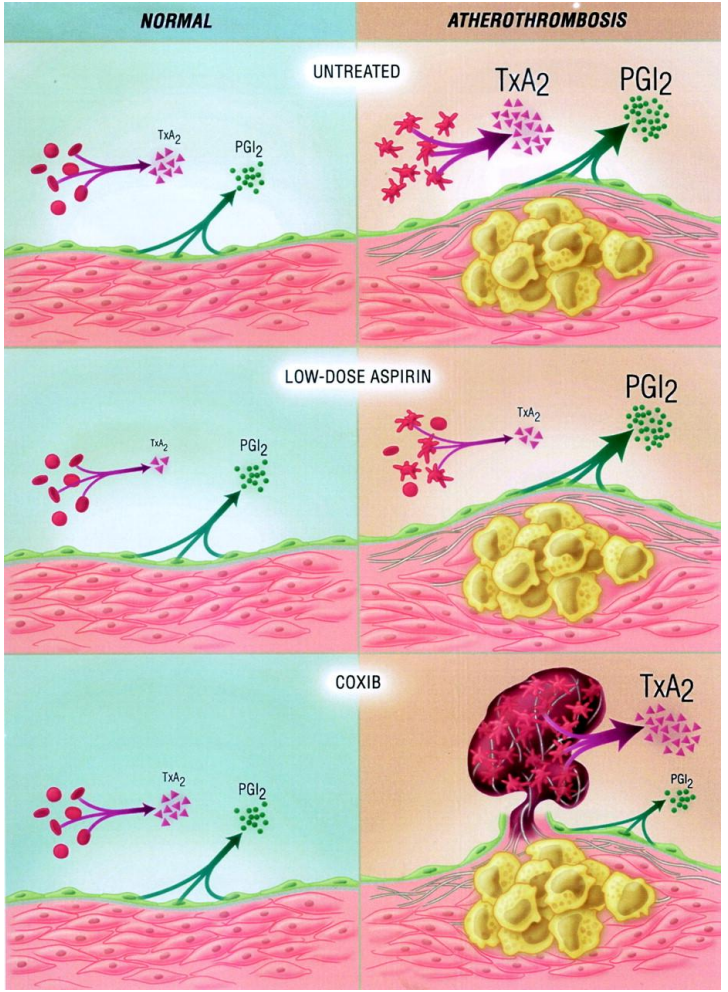
Adverse Effects

Cardiovascular

COX-2 selective NSAIDs improve GI safety but do not inhibit COX-1 formation of TxA2

- prothrombotic and atherogenic

Rofecoxib - withdrawn due to increased incidence of MI, stroke



Adverse Effects

Blood Pressure

5% of NSAID users develop hypertension

Renal

Analgesic nephropathy - slowly progressive renal failure due to chronic NSAID use



Specific NSAIDs

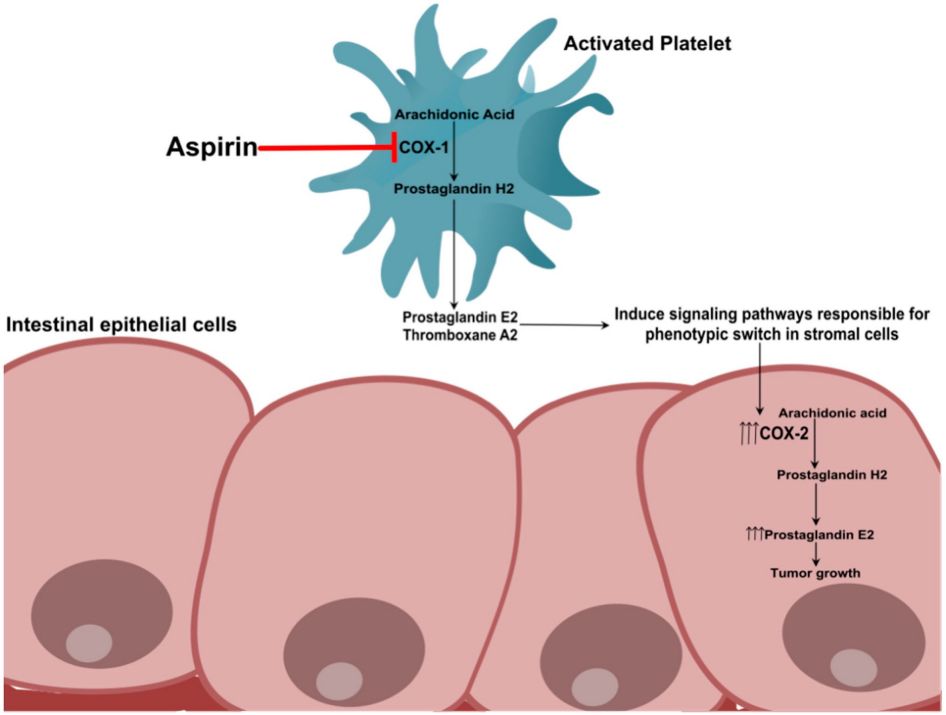
Aspirin (and other salicylates)

Salicylates

- aspirin, salicylic acid, methyl salicylate, sulfasalazine, mesalamine

Mechanism of Action

- **irreversible** inhibition of cyclooxygenase



Aspirin (and other salicylates)

Uses:

1. Analgesic-antipyretic
   - **325-1000mg orally every 4-6hrs**
2. Inflammatory bowel disease
   - mesalamine
3. Rheumatoid arthritis
   - sulfasalazine
4. Warts, corns, fungal infections
   - salicylic acid




6 PLASTERS
SALICYLIC ACID
MEDIPLAST
Calluses, Corns and Warts Plaster
Keratolytic
Distributed by:
PHILUSA CORPORATION
6 PLASTERS
SALICYLIC ACID
MEDIPLAST
Calluses, Corns and Warts Plaster
Keratolytic
SALICYLIC ACID
MEDIPLAST Plaster
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EXFOLIATE
PAULA'S CHOICE
SKIN PERFECTING
2% BHA Liquid Exfoliant
SALICYLIC ACID
All Skin Types
UNCLOGS & SHRINKS ENLARGED PORES
SMOOTHS & EVENS SKIN TONE
LIGHTWEIGHT LIQUID ABSORBS QUICKLY
118 ml / 4 fl. oz.
LA ROCHE-POSAY
LABORATOIRE DERMATOLOGIQUE
EFFACLAR
SALICYLIC ACID
ACNE TREATMENT
WITH LA ROCHE-POSAY THERMAL SPRING WATER
AGE bright clearing serum
dermalogica
Active Clearing
1 US FL OZ / 30 mL

Acetaminophen

Raises threshold to painful stimuli, thus exerting an analgesic effect

Available without prescription, common household analgesic

Combined with other analgesics (opioid and nonopioid)



Acetaminophen

Mechanism of action

- nonselective COX inhibitor but only weak anti-inflammatory effect

Typical dose is **325-650mg every 4-6 hours**

Total daily dose should not exceed **4g** (2g for chronic alcoholics)




PHENYLEPHRINE HCl
PARACETAMOL
100 TABLETS
Neozep® Non-Drowsy
10 mg / 500 mg Tablet
• Nasal Decongestant
• Analgesic-Antipyretic


FEVER
BODY ACHES
SEVERE COLDS
PHENYLEPHRINE HCl
CHLORPHENAMINE MALEATE
PARACETAMOL
BIOFLU®
10 mg / 2 mg / 500 mg TABLET
• Nasal Decongestant • Antihistamine • Analgesic-Antipyretic
5 tablets to go


DEXTROMETHORPHAN HBr
PHENYLPROPANOLAMINE HCl
PARACETAMOL
TUSERAN® FORTE
15 mg / 25 mg / 325 mg capsule
Reformulated
100 capsules
TUSERAN FORTE
COUGH & COLD SOLUTION
ANTITUSSIVE • NASAL DECONGESTANT
ANALGESIC - ANTIPYRETIC
STOPS COUGH
RELIEVES COLDS


PARACETAMOL
CAFFEINE
REXIDOL® FORTE
500 mg / 65 mg Tablet
100
Tablets

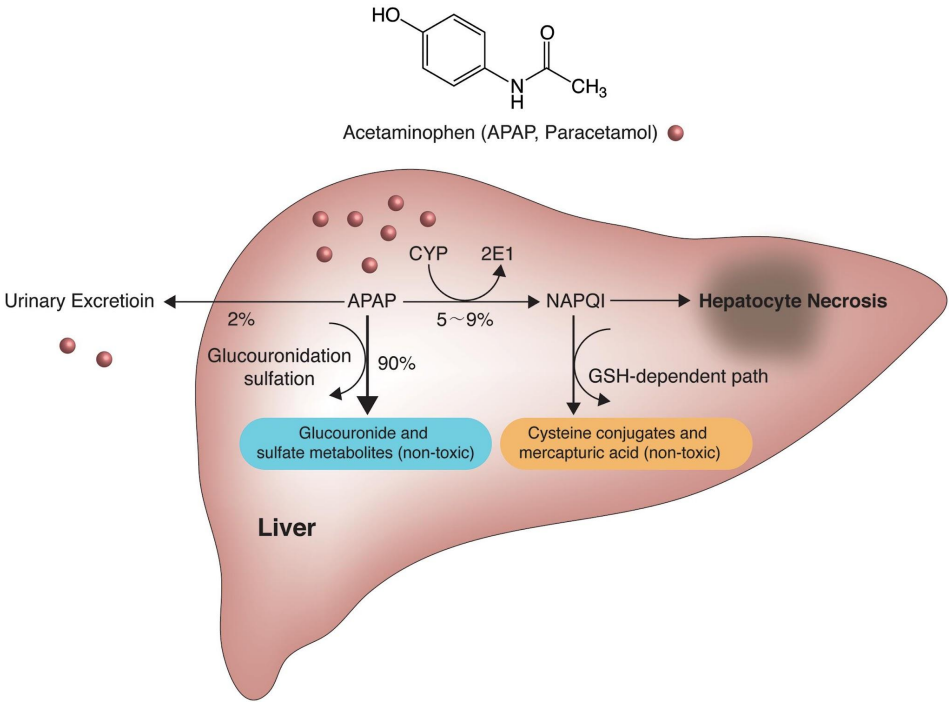
Acetaminophen

Toxicity

Gi adverse effects are less common than other NSAIDs

Hepatotoxicity occur after an ingestion of a **single dose of 10-15g**

20-25g or more is potentially fatal



Diclofenac

Potency is substantially greater than other NSAIDs

Uses: long-term symptomatic treatment of arthritis, primary dysmenorrhea, acute migraine

Dose: **50-150mg** divided in doses



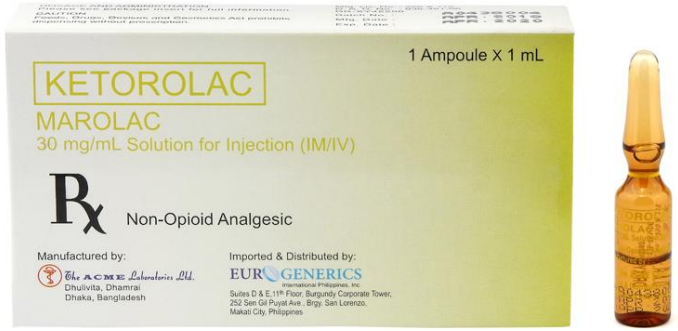
Ketorolac

Potent analgesic but only moderately effective anti-inflammatory drug

Oral (10-20mg)

IV (15-30mg)

IM (30-60mg)



EUROGENERICS
INTERNATIONAL PHILIPPINES INC.

Propionic Acid Derivatives

Ibuprofen, Naproxen, Ketoprofen

Mechanism of Action:

- Nonselective COX inhibitors
- Naproxen has leukocyte inhibitory functions

Ibuprofen

Oral doses containing 200mg or less are available without prescription



Usual dose for mild to moderate pain is **400mg every 4-6hrs** as needed



Naproxen


100
TABLETS
UNILAB
NAPROXEN SODIUM
PARACETAMOL
SKELAN® Forte
275 mg / 325 mg FILM-COATED TABLET
Non-steroidal Anti-inflammatory Drug
Analgesic - Antipyretic
Rx


500 Tablets
Naproxen Sodium
Flanax® 275 mg
Tablet
Non-steroidal
Manufactured by
Interphil Laboratories, Inc.
50 Tablets
Naproxen Sodium
Flanax® Forte
550 mg Tablet
Non-steroidal Anti-inflammatory/Analgesic
Manufactured by
Interphil Laboratories, Inc.
Canlubang Industrial Estate
Bo. Pittland, Cabuyao
Laguna, Philippines for
Taisho Pharmaceuticals (Philippines), Inc.
8th Floor CLIPP Center Building, 11th Ave.
cor. 39th St., Bonifacio Global City, Taguig
Distributed in the Philippines by
ZUELLIG PHARMA CORP.
Km. 14 West Service Road South Superhighway
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TAISHO

Fenamates

Mefenamic acid, meclofenamate, flufenamic acid



No advantages over other NSAIDs



Not recommended for children or pregnant women

COX-2 Inhibitors

Celecoxib, Etoricoxib

Avoided in patients prone to cardiovascular or cerebrovascular disease

Reduces GI complications



Thank you!